

Is anchoring at 'dead' a theoretical requirement for health state valuation?

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Abstract

Values that accompany generic health measures are typically anchored at 1 = full health and 0 = dead. Some health states may then be considered 'worse than dead' (WTD) and assigned negative values, which causes fundamental measurement problems. In this paper, we challenge the assumption that anchoring values at 'dead = 0' is necessary for quality-adjusted life year (QALY) estimation. We summarise the role of 'dead' in health state valuation and consider three critical questions: (i) whether the measurement properties of health state values require 'dead'; (ii) whether 'dead' needs to be valued relative to health states; and (iii) whether values for states WTD are meaningful or useful. We conclude that anchoring 0 at dead is not a requirement of health status measurement or cost-effectiveness analysis. This results from reframing QALYs as the relevant unit of measurement and reframing values as being derived from QALYs rather than the reverse.

KEYWORDS

dead, death, EQ-5D, health-related quality of life, health state valuation, quality-adjusted life year, stated preference, time trade-off, TTO, utility

JEL CLASSIFICATION

I10, J17

1 | INTRODUCTION

Quality-adjusted life year (QALY) calculation has widely adopted methodological conventions. Single-number scores, called here 'health state values' (HSVs), accompany health states described by instruments such as the EQ-5D. HSVs are measured on a scale anchored at 1 meaning 'full health', and 0 meaning 'dead' or a health state equivalent in value to 'dead'. Health states can take negative values, labelled 'worse than dead' (WTD), and stated preference methods must therefore be capable of eliciting both positive ('better than dead': BTDD) and negative values.

Problems with defining 'dead' as a health state are well-known (Kind & Rosser, 1979; Macran & Kind, 2001). Anchoring at dead and estimating WTD values adds considerable complexity to health state valuation studies (Augestad et al., 2013; Bernfort et al., 2018; Little et al., 2014; Parkin, 2011; Schneider, 2022). Devising consistent methods for

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valuing BTd and WTD states is challenging, and research has not identified a satisfactory solution. Elicitation and modelling of WTD values affect the estimated HSVs for all states, not just those WTD.

We have elsewhere reviewed the problems arising from including 'dead' in health state valuation (Sampson et al., 2020). Here, we summarise and develop our argument, proposing a new interpretation of the relationship between HSVs and QALYs. We conclude that it is not necessary to specify a meaning for a state with HSV = 0; estimating HSVs does not rely on identifying a 'dead' state; and it is neither necessary nor useful to distinguish between states BTd or WTD.

2 | THE ROLE OF 'DEAD' IN HEALTH STATE VALUATION

Mortality has been used to measure population health for centuries, and was retained as a component of the more ambitious health measures developed by the 1960s health status indicator movement. For example, the influential paper by Fanshel and Bush (1970), which developed a valuation-based health state index described in terms of function and dysfunction, included death as "a state of total dysfunction", such that its HSV = 0. The equally influential paper by Torrance et al. (1972) placed index weighting in the context of utility measurement, retaining the assumption that 'the healthy state' has a value = 1 and 'the dead state' has a value = 0. This more general concept of health status introduced the possibility of negative health state values, interpreted as states "worse than death". 'Dead' therefore was not simply a scale anchor, but took an active role in valuation. Effectively, states BTd are valued compared with the preferred state of full health, and states WTD with the preferred state of dead.

This approach, with some modifications, has become dominant in valuation studies using the time trade-off (TTO) method, where respondents choose between two 'lives', one in 'full health', the other an impaired health state for a longer duration. 'Dead' has two roles. First, respondents are usually told to assume that death follows each alternative. Second, where respondents are willing to trade off all time in full health for some health states, the interpretation is that they regard a life spent in that health state as worse than being dead; this interpretation is sometimes tested by asking respondents to choose between life in that health state and 'immediate death' (Dolan et al., 1994). Respondents then face different elicitation tasks for WTD and BTd states.

TTO variants, such as lead and lag time, elicit both positive and negative values using a single task (Devlin et al., 2013; Lugnér & Krabbe, 2020) without identifying these as BTd and WTD. However, the EuroQol Group's protocol for EQ-5D valuation studies requires that respondents identify health states as BTd or WTD and uses conventional TTO for BTd and lead time TTO for WTD (Stolk et al., 2022).

3 | PROBLEMS WITH USING 'DEAD'

Including 'dead' in a health state value scale is contestable because 'dead' is not a health state. Health attributes used to compare health states cannot meaningfully describe being dead. A dead person feels no pain, but this is different to saying that a living person feels no pain; a dead person cannot feel pain.

People can use their knowledge of illness to judge the stylised health states used in valuation studies, but few could claim to know death or a state WTD. This applies a fortiori to patient- or experience-based valuation studies, which cannot invoke 'dead' as a comparator (Bernfort et al., 2018).

Evidence shows that preferences for 'dead' are dominated by factors other than the value of health (Sharma & Stano, 2010); that when completing valuation tasks people are not always aware of what 'dead' implies or the meaning of WTD states (Al Sayah et al., 2016); that the framing of 'dead' influences valuations (Jakubczyk et al., 2023); and that negative values are not associated with the states' severity (Roudijk et al., 2022).¹

Including 'dead' in valuation studies produces many excluded observations. 36 of 66 studies reviewed by Engel et al. (2016) excluded responses because of their relationship with the 'dead' state, or lack of it.

Different approaches to the use of 'dead' yield substantially different data characteristics, for example, the minimum value that states can take. The Measurement and Valuation of Health study (MVH Group, 1995), whose pioneering protocol was widely adopted in valuing EQ-5D-3L, used a 10 year time horizon with three months as the shortest possible time in full health, generating a minimum value of -39, with a non-linear scale for WTD values. These were re-scaled so that the minimum was -1, reflecting a view that the scale should be symmetrical around zero. By contrast, the EQ-5D-5L value set for England study censored values at -1 (Devlin, Shah, Feng, Mulhern, & van Hout, 2018) and Yang

et al. (2022) censored at 0. Censoring choices can have an important impact on the distribution of negative values (Liao et al., 2022).

These are important issues since including 'dead' affects the values that studies report (Nord, 1991; Sutherland et al., 1983).

4 | CHALLENGING THE NEED FOR 'DEAD'

Do the measurement properties of HSVs require 'dead'? Historically, the measurement scale for HSVs has been treated as independent of its role in calculating QALYs (Pliskin et al., 1980), with research focused on identifying anchor points and scale properties, and an assertion that values should be measured on at least an interval scale. But nothing in scale or utility measurement theory suggests that the scale must be anchored with 'dead' as zero.

Torrance et al. (1972) made a major contribution by casting HSVs as von Neumann-Morgenstern utilities (Von Neumann & Morgenstern, 1944), to be elicited by offering respondents choices between alternatives that have different outcomes and probabilities of outcome. The 'standard gamble' (SG) method simplifies this by specifying one alternative – the health state to be valued – as a certainty, and the other as a gamble that has two possible health state outcomes, valued at one and zero. Which health states are given those 'anchor' values is arbitrary, with no effect on the scale's interval properties. The zero-valued health state does not have to be 'dead'.

Torrance et al. (1972) developed the TTO method to overcome practical difficulties with SG. The TTO assigns a duration to health states rather than a probability. As with SG, any health state could be defined as representing zero. Torrance et al. chose 'dead' as the zero state, but admitted that this was arbitrary and provided no justification. Others have rationalised the use of 'dead' for scale measurement in the context of QALYs (Miyamoto et al., 1998; Roudijk et al., 2018), but this means neither that such use is necessary nor addresses the challenges outlined in Section 3.

Death is a possible outcome of illness (delaying death is one objective of health care), and resource allocation decisions may involve prioritisation between life-extending and life-improving interventions. Is it therefore essential to place a value on being dead comparable to the values attached to health?

The value attributed to 'dead' cannot affect the QALYs that are gained in any context; by definition, no QALYs can be generated by a dead person. Consequently, the concept of a value for 'dead' is meaningless; whatever that value, and whatever the duration over which QALYs are calculated, the number of QALYs is zero.

People may express a preference for 'dead' and voluntarily describe a health state as WTD. However, it does not follow that 'dead' must play a role in health state valuation. People express preferences for many different kinds of states and relate them to their health, especially when asked directly. For instance, people might readily sacrifice health or longevity to be taller, have children, or be published in the *American Economic Review* (Attema et al., 2014).

Are WTD values meaningful and useful? Someone may judge that a health state identified in a valuation study is WTD, but may not take into account how their assumed preferences for death could be realised. Valuation studies ignore such considerations, but real-life decisions involving choosing to die depend on many issues other than the value of health.

QALY-gain calculations involving states WTD may suggest controversial actions. If WTD values have a minimum of -1 , an intervention that improved for a year someone's health from the worst possible state to a state as bad as being dead would generate 1 QALY, the same as an intervention that restored to full health someone who initially had a health state as bad as being dead. QALY maximisers would be indifferent between them. If the minimum is below -1 , the first intervention generates more than 1 QALY. QALY maximisers would prefer the first, so if they are alternatives both people would end up in a state as bad as being dead.

There may be a duty to prioritise those with a state WTD over those in other health states, and to value their gains more highly. However, such arguments apply to any comparisons between two initial health states. States WTD are not obviously a special case for baseline severity arguments.

Distinguishing between BTd and WTD states creates issues that are unsuited for resolution by health state valuation methods. For BTd states, health state preferences can reasonably be expected to be dominated by considerations of health itself. However, preferences for WTD states will also involve the value of being alive. Labelling states as BTd and WTD has no practical use in health care resource allocation decision-making, so the additional challenges and complexities necessary for this distinction are unjustified.

5 | AN ALTERNATIVE APPROACH – QALYS AS THE UNIT OF MEASUREMENT

QALYs are usually conceptualised according to how they are constructed: as the product of a health state's HSV and the time spent in it. TTO is consistent with this, as it standardises values according to a fixed number of years in full health. In effect, any HSV <1 generates, over 1 year, a fraction of a QALY.

The underlying metric of HSVs is ambiguous. Health states are experienced over time, but the HSV does not have a time dimension. We suggest that the relevant foundational metric is the QALY, and that HSVs are simply a convenient means for calculating QALYs. The HSV scale is not a building block for QALYs, but rather the reverse, and the properties required for HSVs are determined by their role in calculating QALYs.

As well as the measurement properties discussed in Section 4, such as whether the values form an interval scale, QALYs have three key features:

1. QALYs measure the *value* of health states over time.
2. QALYs have an inseparable time dimension; the duration over which a health state is experienced.
3. Illness can be represented as a *shortfall* of QALYs over time.

The first feature represents the core meaning for the units in which quality of life is measured, so that QALYs might be more accurately described as representing the value of a healthy year (VHY). We use the term VHY to describe the value that QALYs measure; we do not reject the QALY metric itself.

The second feature defines the units in which the VHY or QALY is measured. One QALY is defined as being in full health for 1 year. The HSV of 'full health' must therefore be 1. This principle extends to all quantities of QALYs and the HSVs of their corresponding health states. For example, if a health state generates 0.5 QALYs over 1 year, its HSV must be 0.5 (or 0.5 of the VHY). The quantity of QALYs experienced therefore determines HSVs, rather than the usual interpretation that QALYs are an accumulation over time of the HSV for a particular health state.

The third feature means that an HSV can equivalently be defined by reference to (i) the quantity of QALYs experienced over time and (ii) a shortfall in the quantity of QALYs compared with full health. For example, a health state valued at 0.7 is equivalently defined as generating 0.7 QALYs over 1 year and a QALY shortfall of 0.3 over 1 year. Illness can be considered as generating less than one VHY over a year or a VHY shortfall over time.

These features give meaning to the concept of negative HSVs. A very poor health state may generate a quality of life shortfall over a year that is valued greater than 1 VHY. Such a state will have a negative HSV. A shortfall of 1.1 VHYS over 1 year implies an HSV of -0.1 . 'Negative quality of life' has no natural meaning, but a *shortfall* in quality of life with a *value* greater than that of a year in perfect health is conceptually meaningful.

This gives meaning to HSV = 0, which is the HSV for a health state experienced over 1 year that generates a shortfall value = 1 VHY (or 1 QALY). The functional characteristics of this state do not matter; it is not necessary to describe it. Conventional health state valuation research does not seek to identify a health state corresponding to every possible HSV. The objective is the reverse, to identify an HSV (or QALY decrement) for every health state that can be described, including those from a health-related quality of life instrument.

It does not matter if there is no descriptor for a health state valued at 0. Apart from 'full health' = 1, which is the anchor that generates 1 VHY over a year, it is not necessary to attach health states to numbers. It would be impossible to do so; language is not flexible enough to create health state descriptors that precisely describe every HSV. Zero is simply a value lying between the worst identified HSV above zero and the best identified HSV below zero.

This argument demonstrates that 'being dead' plays no part in describing health states for valuation purposes. 'Being dead' generates a QALY shortfall = 1 VHY over a year. This is not because it has an HSV = 0 but because it makes it impossible to generate any QALYs. It is therefore incorrect to describe the health state generating a QALY shortfall = 1 VHY as 'being dead' or even 'a health state of equivalent value to being dead'. Since 'being dead' plays no part in the definition of an HSV = 0, health states with HSVs above and below zero should not be labelled as BTM and WTD.

6 | DISCUSSION

Anchoring health state values at 'dead' originated in health measures that viewed health as a functional concept, rather than quality of life, and as a practical solution to defining a scale with desirable properties. It then became a convention.

We present evidence and a rationale for asserting that (i) anchoring at 'dead' is unnecessary, (ii) anchoring at 'dead' is problematic, and (iii) HSVs can be interpreted without reference to 'dead'.

Removing 'dead' from the definition of HSVs changes how we should speak and write about HSVs. The distinction between and reference to states as better or worse than dead should be banished from health state valuation. BTD and WTD hold meaning outside of QALY estimation, such as in the context of physician-assisted dying; their banishment from health state valuation allows for more careful and meaningful usage.

Our proposal implies that differences in HSVs arising from including 'dead' as an anchor should be interpreted as bias; peoples' health state preferences will be biased by the consideration of 'death'. This affects all HSVs and could be avoided by excluding references to 'dead'.

What are the implications for future research? We have two suggestions.

First, there is a need to explore new approaches to valuation. Excluding 'dead' does not solve the challenge of calculating HSVs for health states that generate VHY shortfalls >1 , with negative values. TTO choices cannot offer a shorter time period in full health that would generate such a shortfall. Methods for calculating HSVs for health states that generate a VHY shortfall >1 cannot offer alternatives over concurrent time periods. One option may be to rely on a sequence of periods, as in a chained TTO, some generating VHY shortfalls >1 because of poor health and others generating QALYs in full health at non-concurrent time periods. Full health may occur before ill-health ('lead time') or after ('lag time'). Alternatively, a valuation task may involve the identification of shortfalls relative to a hypothetical life expectancy. In all cases, the challenges are making the scenarios plausible for respondents and accounting for time preference, less of a concern when considering choices over concurrent time periods. Useful lessons may draw not only on experience with mainstream methods, but other approaches, such as kaizen tasks (Craig et al., 2022).

Second, in light of our reinterpretation, there is a need to assess the importance of negative states. It is not known how commonly states would be associated with a shortfall greater than 1 VHY, how frequently these states would be observed in practice, and how relevant these states are to health technology assessment. These characteristics may differ substantially from current observations for WTD states. The importance of negative states should inform the level of effort dedicated to accurately modelling negative values.

Because most valuation studies have been conducted using the EQ-5D, we have used them as an example. However, our analysis applies to any patient-reported outcome measure or set of HSVs used to estimate QALYs.

'Dead' should play no part in valuing health states. Negative HSVs can provide meaningful information about health states but should not be described as 'WTD'. Researchers should abandon the search for a state for which the HSV is equal to zero. QALYs are the metric of value, representing the value of a healthy year, which has no meaningful relation to the value of dying or being dead.

Researchers should therefore 'drop dead', accept that existing approaches are wrong and outdated, and refocus on developing new methods that recognise that QALYs are the driver of health state values rather than vice versa.

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CONFLICT OF INTEREST STATEMENT

The authors have no conflicts of interest relevant to this work.

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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ENDNOTE

¹ For an extended discussion of this evidence, see Sampson et al. (2020).

REFERENCES

- Al Sayah, F., Mladenovic, A., Gaebel, K., Xie, F., & Johnson, J. A. (2016). How dead is dead? Qualitative findings from participants of combined traditional and lead-time time trade-off valuations. *Quality of Life Research*, 25(1), 35–43. <https://doi.org/10.1007/s11136-015-1073-9>
- Attema, A. E., Brouwer, W. B. f., & Van Exel, J. (2014). Your right arm for a publication in AER? *Economic Inquiry*, 52(1), 495–502. <https://doi.org/10.1111/ecin.12013>
- Augestad, L. A., Rand-Hendriksen, K., Stavem, K., & Kristiansen, I. S. (2013). Time trade-off and attitudes toward euthanasia: Implications of using 'death' as an anchor in health state valuation. *Quality of Life Research*, 22(4), 705–714. <https://doi.org/10.1007/s11136-012-0192-9>
- Bernfort, L., Gerdle, B., Husberg, M., & Levin, L.-Å. (2018). People in states worse than dead according to the EQ-5D UK value set: Would they rather be dead? *Quality of Life Research*, 27(7), 1827–1833. <https://doi.org/10.1007/s11136-018-1848-x>
- Craig, B. M., Rand, K., & Hartman, J. D. (2022). Preference paths and their kaizen tasks for small samples. *The Patient - Patient-Centered Outcomes Research*, 15(2), 187–196. <https://doi.org/10.1007/s40271-021-00541-z>
- Devlin, N., Buckingham, K., Shah, K., Tsuchiya, A., Tilling, C., Wilkinson, G., & van Hout, B. (2013). A comparison of alternative variants of the lead and lag time TTO. *Health Economics*, 22(5), 517–532. <https://doi.org/10.1002/hec.2819>
- Devlin, N. J., Shah, K. K., Feng, Y., Mulhern, B., & van Hout, B. (2018). Valuing health-related quality of life: An EQ-5D-5L value set for England. *Health Economics*, 27(1), 7–22. <https://doi.org/10.1002/hec.3564>
- Dolan, P., Gudex, C., Kind, P., & Williams, A. (1994). *The measurement and valuation of health: First report on the main survey*. Centre for Health Economics, University of York.
- Engel, L., Bansback, N., Bryan, S., Doyle-Waters, M. M., & Whitehurst, D. G. T. (2016). Exclusion criteria in national health state valuation studies: A systematic review. *Medical Decision Making*, 36(7), 798–810. <https://doi.org/10.1177/0272989X15595365>
- Fanshel, S., & Bush, J. W. (1970). A health-status index and its application to health-services outcomes. *Operations Research*, 18(6), 1021–1066. <https://doi.org/10.1287/opre.18.6.1021>
- Jakubczyk, M., Schneider, P., Lipman, S. A., & Sampson, C. (2023). This dead or that dead: Framing effects in the evaluation of health states. *Value in Health*, 27(1), 95–103. <https://doi.org/10.1016/j.jval.2023.10.009>
- Kind, P., & Rosser, R. (1979). Death and dying: Scaling of death for health status indices. In B. Barber, F. Grémy, K. Überla, & G. Wagner (Eds.), *Medical informatics Berlin 1979* (Vol. 5, pp. 28–36). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-93120-8_4
- Liao, M., Rand, K., Yang, Z., Hsu, C.-N., Lin, H.-W., & Luo, N. (2022). Censoring in the time trade-off valuation of worse-than-dead EQ-5D-5L health states: Can a time-based willingness-to-accept question be the solution? *Quality of Life Research*, 32(4), 1165–1174. <https://doi.org/10.1007/s11136-022-03329-2>
- Little, M. H. R., Reitmeir, P., Peters, A., & Leidl, R. (2014). The impact of differences between patient and general population EQ-5D-3L values on the mean tariff scores of different patient groups. *Value in Health*, 17(4), 364–371. <https://doi.org/10.1016/j.jval.2014.02.002>
- Lugnér, A. K., & Krabbe, P. F. M. (2020). An overview of the time trade-off method: Concept, foundation, and the evaluation of distorting factors in putting a value on health. *Expert Review of Pharmacoeconomics & Outcomes Research*, 20(4), 331–342. <https://doi.org/10.1080/14737167.2020.1779062>
- Macran, S., & Kind, P. (2001). 'Death' and the valuation of health-related quality of life. *Medical Care*, 39(3), 217–227. <https://doi.org/10.1097/00005650-200103000-00003>
- Miyamoto, J. M., Wakker, P. P., Bleichrodt, H., & Peters, H. J. M. (1998). The zero-condition: A simplifying assumption in QALY measurement and multiattribute utility. *Management Science*, 44(6), 839–849. <https://doi.org/10.1287/mnsc.44.6.839>
- MVH Group. (1995). *The measurement and valuation of health: Final report on the modelling of valuation tariffs*. Centre for Health Economics, University of York.
- Nord, E. (1991). EuroQol: Health-related quality of life measurement. Valuations of health states by the general public in Norway. *Health Policy*, 18(1), 25–36. [https://doi.org/10.1016/0168-8510\(91\)90141-j](https://doi.org/10.1016/0168-8510(91)90141-j)
- Parkin, D. (2011). *What does a health value of zero mean?* [Blog]. The Academic Health Economists' Blog. Retrieved from <https://aheblog.com/2011/07/13/what-does-a-health-value-of-zero-mean/>
- Pliskin, J. S., Shepard, D. S., & Weinstein, M. C. (1980). Utility functions for life years and health status. *Operations Research*, 28(1), 206–224. <https://doi.org/10.1287/opre.28.1.206>
- Roudijk, B., Donders, A. R. T., & Stalmeier, P. F. M. (2018). Setting dead at zero: Applying scale properties to the QALY model. *Medical Decision Making*, 38(6), 627–634. <https://doi.org/10.1177/0272989X18765184>
- Roudijk, B., Donders, R., & Stalmeier, P. (2022). A threshold explanation for the lack of variation in negative composite time trade-off values. *Quality of Life Research*, 31(9), 2753–2761. <https://doi.org/10.1007/s11136-022-03155-6>
- Sampson, C., Parkin, D., & Devlin, N. (2020). *Drop dead: Is anchoring at 'Dead' a theoretical requirement in health state valuation?* Office of Health Economics. Retrieved from <https://www.ohe.org/publications/drop-dead-anchoring-dead-theoretical-requirement-health-state-valuation/>
- Schneider, P. (2022). The QALY is ableist: On the unethical implications of health states worse than dead. *Quality of Life Research*, 31(5), 1545–1552. <https://doi.org/10.1007/s11136-021-03052-4>

- Sharma, R., & Stano, M. (2010). Implications of an economic model of health states worse than dead. *Journal of Health Economics*, 29(4), 536–540. <https://doi.org/10.1016/j.jhealeco.2010.05.005>
- Stolk, E., Ramos-Goñi, J. M., Ludwig, K., Oppe, M., & Norman, R. (2022). The development and strengthening of methods for valuing EQ-5D-5L – An overview. In N. Devlin, B. Roudijk, & K. Ludwig (Eds.), *Value sets for EQ-5D-5L: A compendium, comparative Review & user guide*. Springer. Retrieved from <http://www.ncbi.nlm.nih.gov/books/NBK589298/>
- Sutherland, H. J., Dunn, V., & Boyd, N. F. (1983). Measurement of values for states of health with linear analog scales. *Medical Decision Making*, 3(4), 477–487. <https://doi.org/10.1177/0272989X8300300406>
- Torrance, G. W., Thomas, W. H., & Sackett, D. L. (1972). A utility maximization model for evaluation of health care programs. *Health Services Research*, 7(2), 118–133.
- Von Neumann, J., & Morgenstern, O. (1944). *Theory of games and economic behavior*. Princeton University Press.
- Yang, Z., Rand, K., & Luo, N. (2022). Censoring time trade-off values at 0 versus at –1: How does the assumption for worse-than-dead time trade-off values affect the modeling of EQ-5D-5L valuation data? *Value in Health*, 26(5), 685–693. <https://doi.org/10.1016/j.jval.2022.11.001>

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